

Kanishka Singh

☎ +1-646-781-6532 ✉ kanishka.singh33@gmail.com in singh-kanishka 🌐 up-the-ladder 📁 portfolio

Education

New York University - Courant Institute of Mathematical Sciences M.S. in Computer Science <i>Relevant Courses: Machine Learning, Vision Meets ML, Predictive Analytics</i>	Sept 2025 – Present New York, USA
Kalinga Institute of Industrial Technology B.Tech. in Computer Science & Engineering (GPA: 9.04/10.0)	Sept 2020 – Apr 2024 Bhubaneswar, India

Industry Experience

Ernst & Young, Senior Analyst • Built GenAI-powered chatbots using LangChain and RAG , designing evaluation strategies for response relevance , latency trade-offs , and production monitoring signals in real-world deployments, delivering CXO-level insights. • Resolved ambiguity and contextual understanding challenges in chatbot systems by 90% by implementing LangChain memory modules (ConversationBufferMemory, ConversationSummaryMemory, VectorStoreRetrieverMemory) and designing robust few-shot prompts for improved conversational continuity. • Engineered a Stock-Out Prediction Engine using feature correlation analysis and ML models (Logistic Regression, Decision Trees, Gradient Boosting) on 2+ years of inventory data , achieving 85% accuracy on unseen data and enabling data-driven inventory decisions to mitigate stock-out risk.	Jul 2024 – Apr 2025
Ernst & Young, Data Science & Analytics Intern • Developed “ Dashboard Genie ”, a GenAI-powered KPI discovery assistant leveraging embeddings , vector search , and LangChain to build a semantic retrieval system enabling 15+ managers to search KPIs across enterprise dashboards, surface top-3 relevant matches with direct links, and significantly reduce time-to-insight and decision-making latency. • Served as Technical Lead for an EV dashboard POC for a major automotive client, re-architecting the backend to handle evolving business logic and implementing scalable data pipelines and backend systems for real-time analytics , including ETL workflows , optimized SQL views , and interactive dashboards , delivering end-to-end analytics . • Designed and automated a Pareto Analysis dashboard using Python and SQL , achieving 80% backend automation and streamlining manual analytical workflows .	Jun 2023 – Jul 2024

Research Experience

Sentiment-Aware Bitcoin Volatility Forecasting with Residual Decomposition Networks <i>Prof. Anasse Bari, New York University - Courant Institute of Mathematical Sciences</i>	Sept 2025 – Dec 2025
• Developed an N-BEATS-based volatility forecasting model with a task-aware hybrid loss , achieving MAE 0.2897 and outperforming LSTM, CNN, Transformer, and TFT baselines. • Designed a robust hybrid loss capturing jump sensitivity and long-term trends , enabling strong generalization under heavy-tailed and regime-shifting conditions .	

Projects

Bubble Rush Grand Prix • Built a real-time browser game engine in React using requestAnimationFrame , delta-time normalization , and collision detection , fully decoupling game logic from rendering. • Engineered deterministic spawning , lane-based constraints , and a localStorage leaderboard ; deployed via GitHub Pages with version-controlled workflows.	<i>React, JavaScript, Canvas API, Vite, GitHub Pages</i>
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Skills

Languages: Python, **SQL**, C/C++, JavaScript, TypeScript, Java, R
LLM & AI Systems: Large Language Models (LLMs), Generative AI, RAG Pipelines, LLM Systems, Evaluation Strategies, LangChain, Prompt Engineering
ML & Data: PyTorch, TensorFlow, Scikit-learn, PySpark, **Data Pipelines**, **ETL Workflows**, Feature Engineering, Model Evaluation
Systems & Backend: Backend Development, Scalable Systems, System Design, REST APIs, **Real-time Analytics**, Vector Search, Embeddings
Tools: Git, Docker, Jira, Agile (Scrum, Kanban)